

Complete Linux Server Malware Assessment Guide (Web, Application, Database & Infrastructure Servers)

Linux Persistence Mechanisms



Cron Jobs

Systemd
Services

Shell RC Files

Init.d / rc.local

SSH
Authorized
Keys

LD_Preload
Rootkits

This guide provides detailed malware assessment and persistence analysis for enterprise Red Hat Linux servers hosting Apache Web Server, Apache Tomcat Application Server, MySQL Database Server, and supporting infrastructure services. The document covers suspicious files, malware signatures, attack indicators, forensic paths, detection tools, monitoring commands, and security best practices.

1. Web Server (Apache HTTPD) Malware Indicators

Critical Directories:

- /etc/httpd/
- /var/www/html/
- /var/log/httpd/
- /usr/lib/cgi-bin/

Common Malware & Webshell Indicators:

- Suspicious PHP/JSP upload files
- Encoded eval/base64 payloads
- Webshell names like cmd.jsp, shell.jsp, adminer.php
- Hidden .htaccess redirect malware

- Crypto-mining JavaScript injections

Log Indicators:

- Multiple POST requests to upload endpoints
- Requests with base64 encoded parameters
- Unexpected outbound connections
- Access from TOR/VPN IPs

Detection Commands:

```
find /var/www -name "*.jsp"
grep -Ri "eval|base64_decode|Runtime.getRuntime" /var/www
tail -f /var/log/httpd/access_log
```

2. Application Server (Apache Tomcat) Malware Indicators

Critical Directories:

- /opt/tomcat/webapps/
- /opt/tomcat/conf/
- /opt/tomcat/logs/
- /tmp/

Common Tomcat Malware Techniques:

- Malicious WAR deployments
- JSP webshell uploads
- Memory-resident shells
- Reverse shell execution from manager app
- Credential theft from server.xml

Suspicious Files:

- shell.jsp
- cmd.jsp
- upload.jsp
- ROOT.war with hidden payloads

Indicators of Compromise:

- Unexpected WAR files
- Modified web.xml files
- High CPU usage by java process
- Java processes connecting externally

Detection Commands:

```
ps -ef | grep java
```

```
netstat -plant | grep java  
find webapps -name "*.jsp"  
grep -Ri "ProcessBuilder\|Runtime.getRuntime" webapps/
```

3. Database Server (MySQL/MariaDB) Malware Indicators

Critical Directories:

- /var/lib/mysql/
- /etc/my.cnf
- /var/log/mysqld.log

Attack Techniques:

- Unauthorized database users
- UDF (User Defined Function) backdoors
- SQL injection persistence
- Data exfiltration scripts
- Malware hiding inside scheduled events

Indicators:

- Unknown MySQL users
- Unexpected FILE privileges
- Suspicious shared libraries (.so)
- Large outbound traffic from mysqld

Detection Commands:

```
mysql -e "SELECT user,host FROM mysql.user;"  
mysql -e "SHOW GRANTS;"  
find /var/lib/mysql -name "*.so"  
grep -Ri "event_scheduler" /etc/my.cnf
```

4. SSH & Remote Access Persistence

Persistence Files:

- ~/.ssh/authorized_keys
- /etc/ssh/sshd_config

Indicators:

- Unknown SSH keys
- Root login enabled unexpectedly
- SSH tunneling activity
- Brute-force attempts in auth logs

Detection Commands:

```
last -a  
grep "Failed password" /var/log/secure  
find / -name authorized_keys 2>/dev/null
```

5. Cron Jobs & Scheduled Malware Tasks

Malware commonly schedules itself using cron jobs. **Critical Paths:**

- /etc/crontab
- /var/spool/cron/
- /etc/cron.d/

Indicators:

- wget/curl download commands
- Base64 encoded scripts
- Hidden jobs running every minute

Detection Commands:

```
crontab -l  
grep -R "wget|curl|nc" /etc/cron*
```

6. Rootkits & LD_PRELOAD Malware

Rootkits attempt to hide processes, ports, and malicious files. **Critical Files:**

- /etc/ld.so.preload
- Suspicious .so libraries

Indicators:

- Hidden ports/processes
- Missing files in ls output
- Network anomalies

Detection Tools:

- chkrootkit
- rkhunter
- Lynis
- auditd

Server Comparison Matrix

Server Type	Main Risk	Common Malware	Critical Logs	Priority
Apache Web Server	Webshell Uploads	JSP/PHP Shells	/var/log/httpd/	High
Tomcat Application	WAR Exploits	Memory Shells	catalina.out	Critical
MySQL Database	Privilege Abuse	UDF Backdoors	mysqld.log	Critical
SSH Services	Unauthorized Access	Key Persistence	/var/log/secure	High
Cron/Systemd	Persistence	Scheduled Malware	journalctl	High

Recommended Enterprise Security Tools

Host Security:

- Wazuh
- OSSEC
- AIDE
- auditd
- SELinux

Malware Detection:

- ClamAV
- YARA Rules
- chkrootkit
- rkhunter

Monitoring & SIEM:

- Splunk
- ELK Stack (Elasticsearch, Logstash, Kibana)
- Graylog
- Prometheus + Grafana

Network Detection:

- Suricata
- Zeek
- Wireshark

Best Practices

- Keep Apache, Tomcat, Java, and MySQL patched.
- Disable unused ports and services.
- Restrict WAR deployment permissions.
- Enable SELinux enforcing mode.
- Use MFA and SSH key rotation.

- Monitor outbound traffic continuously.
- Enable centralized log monitoring and SIEM integration.
- Perform regular malware and integrity scans.